Pathfolio: Project Overview and User Stories

Personal job and school application assistant. Version 1.0 project brief.

1. What is Pathfolio?

Pathfolio is a personal tool that helps one person find jobs and school programs, prepare tailored application documents, apply (by email, by form, or by following guided steps), and keep a complete record of every application sent.

The name combines *path* and *portfolio*: the path a person takes through work and study, and the portfolio of documents that carries them along it. It reflects the goal: stop spending hours on repetitive application work and get things done.

Tagline: *Apply smarter. Track everything.*

2. The problem

Applying for jobs and schools is slow and repetitive:

- Every application needs a CV and cover letter adjusted to the specific role or program.
- Every company or school uses a different way to apply: an email address, a web form, or a multi-page portal.
- Required documents and deadlines differ and change every year.
- After 20 or 30 applications, it is easy to forget where you applied, what you sent, and who you have not heard back from.

3. The solution

Pathfolio handles the full cycle in one place:

1. **Find** matching jobs or programs by region and type.
2. **Tailor** the CV and cover letter to each one, without inventing facts.
3. **Apply** using the best available route (email, automated form filling, or a step-by-step guide).
4. **Track** every application so nothing is forgotten.

4. Who it is for

The first and only user is the builder: a software engineer applying for jobs (for example backend or frontend roles) and for school programs, in one or more regions. Because it is a personal tool, it needs no accounts, payments, or public hosting. The design may later be shared with others, so the document is written to be understood by any developer.

5. Goals and non-goals

Goals

- Cut the time to prepare one application from hours to minutes.

- Produce tailored documents that stay truthful to the original CV.
- Support three ways to apply: email, automated form filling, and guided manual steps.
- Keep a permanent, searchable history of all applications.

Non-goals (for version 1)

- No public product, user accounts, or billing.
- No mass-applying to hundreds of jobs. Quality over volume.
- No bypassing CAPTCHAs or breaking a website's terms of service.
- No fully automatic submission of school portal applications that need transcripts, payments, or recommendation letters.

6. Core principles

1. **Never invent facts.** The AI may reword and re-order the user's real experience, but never add skills, jobs, degrees, or dates that are not in the master profile.
2. **Never invent contacts.** Emails, links, deadlines, and required documents must come from the actual job post or official admissions page. If not found, Pathfolio says "not found" and shows the source it checked.
3. **Human approval before sending.** Pathfolio pauses on a review screen before any email is sent or any form is submitted. Auto-apply is allowed, but only after this check.
4. **Everything is recorded.** Every application is saved automatically, whether Pathfolio sent it or the user applied manually.
5. **Start small.** Build the most valuable parts first and add the rest in stages.

7. How Pathfolio works

7.1 Main flow

1. The user chooses **Job** or **School**.
2. The user selects a region and a category (for example Software Engineer, then Backend or Frontend).
3. Pathfolio lists matching openings or programs, with a match score against the user's profile.
4. The user selects the ones to apply to.
5. Pathfolio tailors the CV and cover letter (or school statement) for each one and shows what changed.
6. Pathfolio picks the application route (below).
7. The user reviews and approves.
8. Pathfolio applies or guides the user, then saves the record in the tracker.

7.2 The three application routes

Route	When it is used	What Pathfolio does
A. Email	The posting lists an application email	Writes the email, attaches the tailored documents, and sends it or saves it as a draft

Route	When it is used	What Pathfolio does
B. Online form	The posting links to a web form	Opens the page, fills in details, uploads documents, answers questions, and pauses before Submit
C. Guided steps	No email or simple form is found, or the process is a portal (typical for schools)	Gives a step-by-step guide: which website and page to open, which buttons to follow, and which documents to prepare

8. Features

8.1 Master profile

One place that stores the user's CV, education, work history, projects, skills, links, contact details, and reusable answers (for example work authorization, notice period, languages). Every application is built from this profile.

8.2 Job and school search

- Filter by region, role type, and level.
- Pull listings from job APIs and public company job boards.
- For schools, search programs and read admission requirements from official pages.
- Show a match score and list of missing skills or requirements.

8.3 AI document tailoring

- Rewrite the CV summary, re-order skills, and highlight relevant experience for each posting.
- Generate a cover letter, application email, or school motivation letter.
- Show a "review changes" screen comparing the original and tailored versions.
- Enforce the no-invention rule by checking output against the master profile.

8.4 Application route detection

- Read the posting page and extract the application email or application link.
- Choose route A, B, or C automatically.
- Display the source link so the user can verify.

8.5 Email applications

- Compose and send through Gmail or SMTP, or save as draft.
- Attach the exact tailored files.

8.6 Automated form filling

- Use browser automation to fill common application forms.

- Handle file uploads and standard questions from the master profile.
- Stop before the final Submit for review.
- If something is unclear (salary, visa, unusual question), ask the user instead of guessing.

8.7 Step-by-step guides

- For each posting or program, show numbered steps: where to go, what to click, what to fill.
- List required documents with a ready or not-ready checkbox.
- Include deadlines and fees when found on the official page.

8.8 School document checklist

- Per school: transcripts, certificates, recommendation letters, test scores, passport, statement of purpose, application fee.
- Status and deadline for each item, with reminders.

8.9 Application tracker

Saves, for every application:

- Company or school, and job title or program
- Date applied and how (email, form, manual)
- Link to the posting
- The exact CV and cover letter versions sent
- Contact email, if any
- Status: saved, applied, interview, rejected, offer, no reply
- Deadline
- Notes (people spoken to, interview dates)

Tracker tools:

- **Duplicate warning** before applying to the same company or role twice.
- **Search and filter** by company, status, or date (for example, "no reply after 14 days").
- **Follow-up reminders** after one or two weeks without reply.
- **Inbox matching (later)** to detect replies and update status.
- **Export** to CSV as a backup.

8.10 Interview preparation (later)

Generate likely questions and talking points for each job based on the posting and the tailored CV.

9. User stories

The user is called "I" below because the first user is the owner of the tool.

Profile

- **US-01.** As a job seeker, I want to upload my CV once and store it as a master profile, so I do not re-enter my details for every application. *Done when:* my CV is parsed into editable sections and saved.

- **US-02.** As a job seeker, I want to store reusable answers (notice period, work authorization, languages), so forms can be filled consistently. *Done when:* I can add, edit, and delete answers.

Search

- **US-03.** As a job seeker, I want to choose a region and a role type (for example backend or frontend), so I only see relevant jobs. *Done when:* results match the chosen filters.
- **US-04.** As a job seeker, I want a match score and a list of missing skills for each job, so I can decide where to apply. *Done when:* each result shows a score and gaps.
- **US-05.** As a student, I want to search programs and see their admission requirements from the official page, so I know what is needed. *Done when:* each program shows requirements with a source link.

Tailoring

- **US-06.** As an applicant, I want my CV tailored to a specific job, so it highlights what that employer wants. *Done when:* a tailored CV is produced from the master profile only.
- **US-07.** As an applicant, I want to see exactly what the AI changed, so I can make sure nothing false was added. *Done when:* a side-by-side comparison is shown and I can accept or reject each change.
- **US-08.** As an applicant, I want a cover letter or motivation letter generated for each application, so I do not write each from scratch. *Done when:* a draft is generated and editable.

Applying

- **US-09.** As an applicant, I want Pathfolio to find the real application email in the job post, so I send my CV to the right place. *Done when:* the email is extracted from the page and shown with its source, or "not found" is displayed.
- **US-10.** As an applicant, I want Pathfolio to send the application email with my tailored documents attached, so I save time. *Done when:* I approve a preview and the email is sent or saved as a draft.
- **US-11.** As an applicant, I want Pathfolio to fill online application forms for me, so I do not type the same details repeatedly. *Done when:* fields and uploads are completed and Pathfolio pauses before Submit.
- **US-12.** As an applicant, I want Pathfolio to ask me when it is unsure about an answer, so it never guesses on important questions. *Done when:* unknown or sensitive questions are flagged for my input.
- **US-13.** As an applicant, I want step-by-step instructions when no email or form is available, so I know which page to open and what to prepare. *Done when:* a numbered guide with links and a documents list is produced from official sources.

Schools

- **US-14.** As a student, I want a checklist of required documents per school with deadlines, so I do not miss anything. *Done when:* each school has items with status and due dates.
- **US-15.** As a student, I want reminders before deadlines, so I submit on time. *Done when:* reminders appear at configured intervals before each deadline.

Tracking

- **US-16.** As an applicant, I want every application saved automatically, so I always know where I applied. *Done when:* each send, submit, or manual entry creates a record.
- **US-17.** As an applicant, I want to be warned if I already applied to the same company or role, so I do not apply twice. *Done when:* a warning appears before applying to a duplicate.
- **US-18.** As an applicant, I want to see which exact CV and cover letter I sent, so I can prepare for interviews. *Done when:* the sent files are stored and openable from the record.
- **US-19.** As an applicant, I want to filter applications by status and age (for example, no reply after 14 days), so I know whom to follow up with. *Done when:* filters work and return correct results.
- **US-20.** As an applicant, I want follow-up reminders, so I do not forget to check in. *Done when:* a reminder is created automatically after a set number of days.
- **US-21.** As an applicant, I want to export all my applications to a spreadsheet, so I have a backup. *Done when:* a CSV file with all records can be downloaded.
- **US-22 (later).** As an applicant, I want Pathfolio to detect replies in my inbox and update statuses, so the tracker stays current without effort. *Done when:* replies from tracked companies update the status after my confirmation.

Interview preparation (later)

- **US-23.** As an applicant, I want likely interview questions for each job, so I can prepare. *Done when:* a list of questions is generated from the posting and my CV.

10. Data model (simplified)

Table	Main fields
profile	name, email, phone, links, summary, skills, education, experience, projects, saved answers
opportunity	type (job or school), title, organization, region, link, deadline, requirements, source
application	opportunity, date applied, method, status, contact email, notes, follow-up date
document	application, type (CV, cover letter, other), file path, version
checklist_item	application or school, document name, status, due date
reminder	application, date, message, done

11. Technical approach

Part	Choice
Language	JavaScript (Node.js), TypeScript optional
AI	LLM API for tailoring, summarizing, and writing
Form filling	Playwright
Reading web pages	Cheerio
Email	Nodemailer or Gmail API
Database	SQLite (better-sqlite3)
CV files	pdf-parse, mammoth, docx
Interface	Command line first, then a simple local web page

Job data sources: public job APIs (such as Adzuna, Jooble, Remotive), and the public job board interfaces offered by Greenhouse and Lever. Scraping large job sites is avoided because it is fragile and may violate their terms. Coverage differs per country, so regional availability must be checked early.

12. Risks and how to handle them

Risk	Handling
AI invents emails or requirements	Only extract from fetched pages; show source; otherwise show "not found"
AI adds false claims to the CV	Generate only from the master profile; show a change review screen
Wrong answer in a form (salary, visa)	Ask the user; pause before Submit
Sites block automation or use CAPTCHAs	Prefer company career pages and email; fall back to guided steps
Platform terms of service	Avoid automating sites that forbid it, such as LinkedIn
Outdated school information	Always read from the current official page and link to it
Data loss	Regular export and backup of the SQLite file
Private data exposure	Keep all data local; protect API keys and email credentials

13. Roadmap

Phase	Deliverable
1	Master profile, paste a job description, get tailored CV and cover letter
2	Application tracker with duplicate warning and export
3	Job search by region and role type, with match score
4	Application email extraction and sending
5	Step-by-step guides and school document checklist
6	Playwright form filling with review before submit
7	School search, deadline reminders, inbox matching, interview prep

14. Success measures

- Time to prepare a tailored application is under 10 minutes.
- Zero invented facts or contacts in generated documents.
- 100% of applications appear in the tracker.
- No duplicate applications sent by accident.
- No missed school deadlines.

15. Open questions

- Which regions and countries should be supported first?
- Which job APIs have the best coverage for those regions?
- Which email provider will be used for sending (Gmail or other)?
- Which schools or programs are the first targets?
- Should the interface stay command line, or move to a local web page early?

16. Name notes

Pathfolio is easy to say and covers both career and study applications. Alternative names considered: *Inzira* ("path" in Kinyarwanda), *Saba* ("to apply" or "to ask"), *Intambwe* ("step" or "progress"), *Dossier*, and *Launchpad*. Before using any name publicly, check that it is not already taken as a domain or trademark.